Social Network Analysis

LINK ANALYSIS

SimRank: Measuring Similarity of Objects

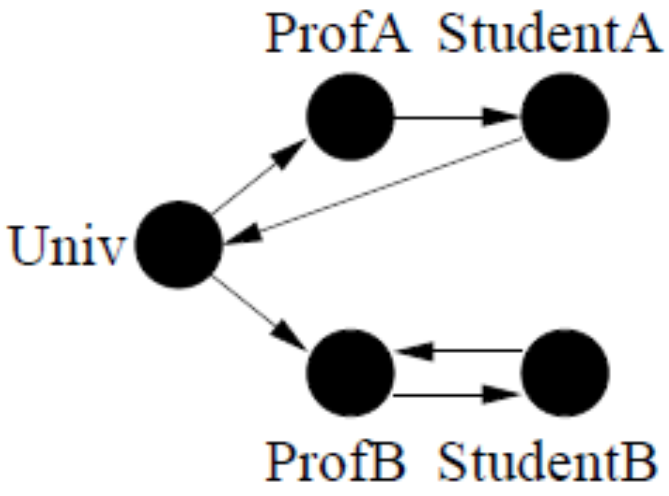
SimRank: Measuring Similarity of Objects

- SimRank measures similarity of the structural context in which objects occur, based on their relationships with other objects.
- Idea : Two objects are similar if they are related to similar objects
- For a given domain, SimRank can be combined with other domain-specific similarity measures.
- Example 1 : Citation graph
  - Two papers are similar if they are cited by similar papers
- Example 2 : E- Commerce graph
  - Two products are similar if they are bought by similar customers
  - Two customers are similar if they are buying similar products

SimRank: Measuring Similarity of Objects

- Idea : Two objects are similar if they are related to similar objects
- More precisely, objects a and b are similar if they are related to objects c and d, respectively, and c and d are themselves similar.
- The base case is that objects are similar to themselves.

SimRank: Measuring Similarity of Objects

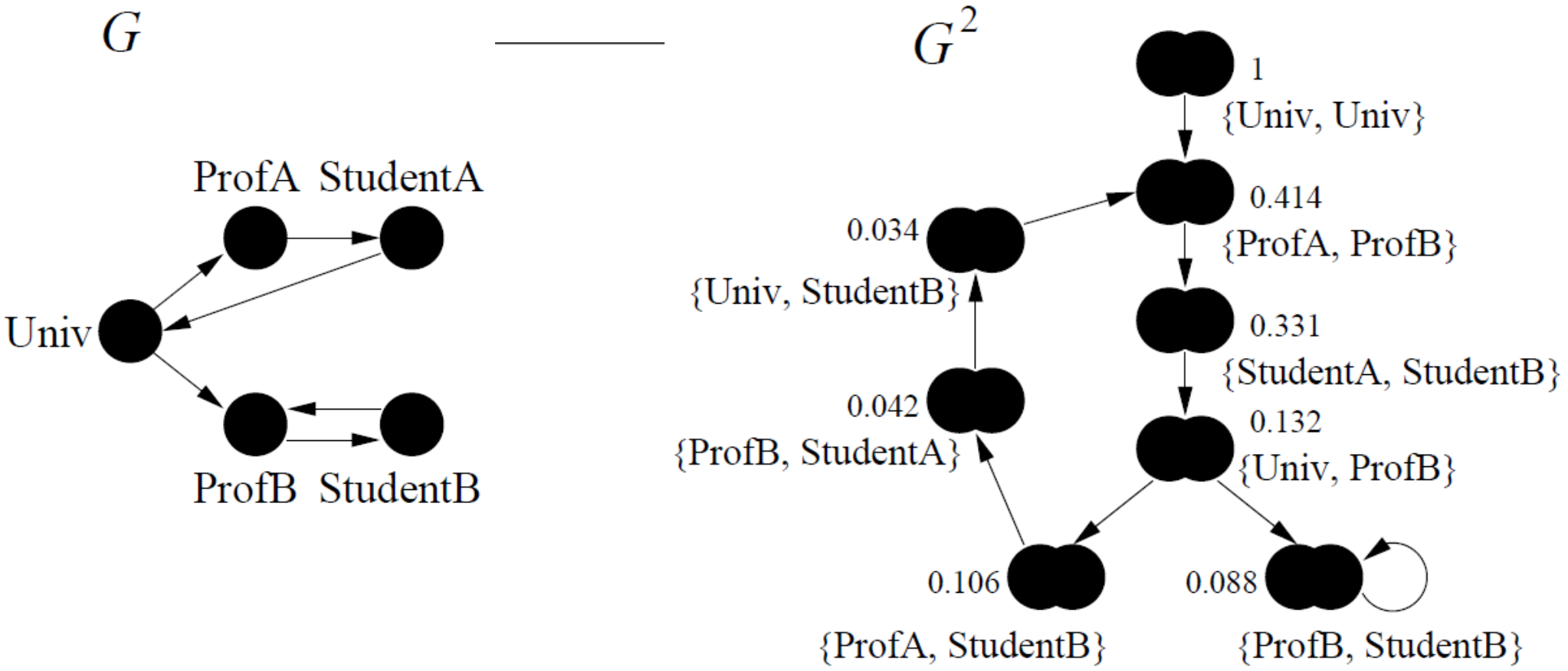


- Graph shows the Web pages of two professors ProfA and ProfB, their students StudentA and StudentB, and the home page of their university Univ.
- Edges between nodes represent hyperlinks from one page to another.
- From the fact that both are referenced (linked to) by Univ, we may infer that ProfA and ProfB are similar
- Can we infer that StudentA and StudentB are also similar based on the similarity of ProfA and ProfB ?
- Similar inference can be derived for other pairs of objects

SimRank: Measuring Similarity of Objects

- Logical representation of the SimRank computation by using a node-pair
- A new graph G^2 is formed in which each node represents an ordered pair of nodes of G.
- A node (a, b) of G^2 points to a node (c, d) if, in G, a points to c and b points to d.
- Each node-pair shows the similarity score between two nodes that they represent
- Scores are symmetric
- Draw (a, b) and (b, a) as a single node {a, b} (with the union of their associated edges).
- Iterative computation of SimRank scores for each node in G^2

SimRank: Measuring Similarity of Objects



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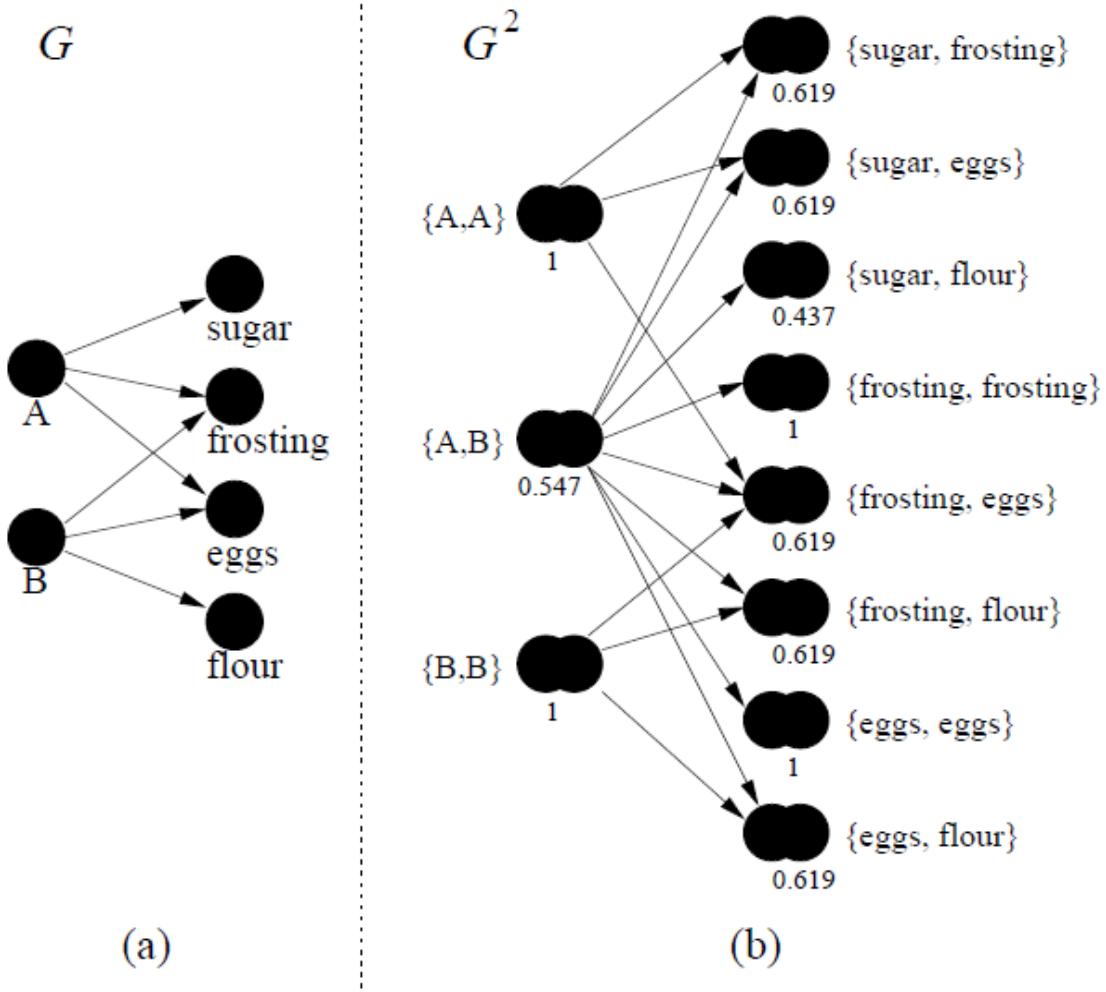


SimRank: Basic Formulation

- For a node v in the network, $I(v) = \{I_i(v) | 1 \leq i \leq |I(v)|\}$ and $O(v) = \{O_i(v) | 1 \leq i \leq |O(v)|\}$ denotes the sets of indegree and outdegree neighbours, respectively.
- Formulate the similarity score $s(a, b) \in [0,1]$ as follows:

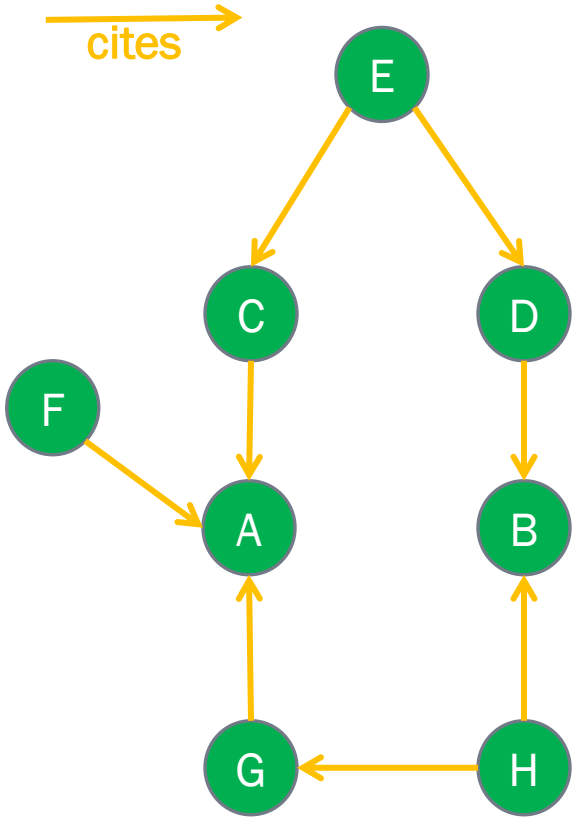
$$s(a,b) = \begin{cases} 1 & if\ a = b \\ 0 & if\ I(a) = \emptyset\ or\ I(b) = \emptyset \\ \dfrac{C}{|I(a)|.|I(b)|} \displaystyle\sum_{i=1}^{|I(a)|} \sum_{j=1}^{|I(b)|} s(I_i(a), I_j(b)) & otherwise \end{cases}$$

  - A node is maximally similar to itself
  - No way of determining the score for a neighborhood that does not exist
  - Similarity between two randomly selected nodes is proportional to the average similarity between their neighbors

SimRank: Basic Formulation

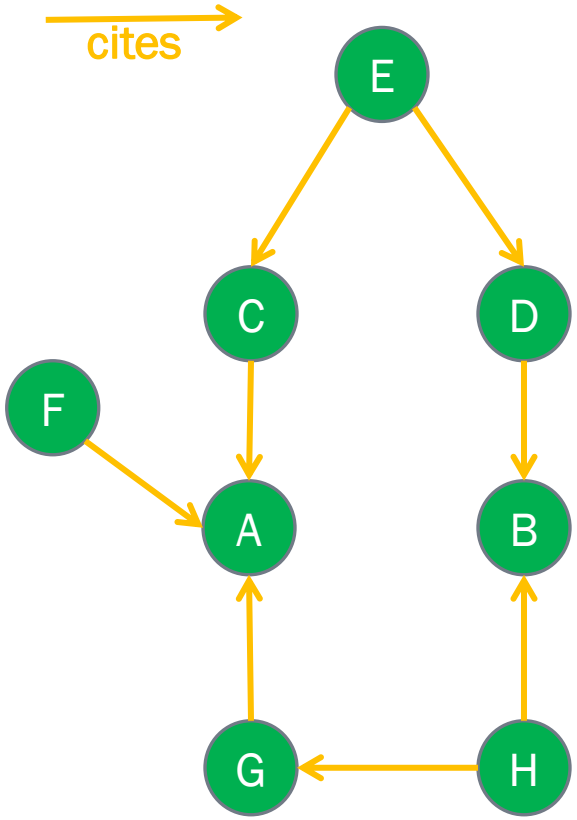
- **Constant C** is considered as a confidence level or a decay factor
- Consider a simple scenario where page x references both c and d, so we conclude some similarity between c and d.
- The similarity of x with itself is 1, but we probably don't want to conclude that s(c, d) = s(x, x) = 1.
- Rather, we let s(c, d) = C x s(x, x), meaning that we are less confident about the similarity between c and d than we are between x and itself

SimRank: Example 2 Citation Network



- Paper E cites papers C and D
  - Papers C and D appears similar
- Paper H cites papers B and G
  - Papers B and G appears similar
- What about the similarity of papers A and B?
- Lets calculate…

SimRank: Example 2 Citation Network



In-neighbors:

- I(A)={C,F,G} I(B)={D,H}
- s(A,B)=0.8/6 (s(C,D)+s(C,H)+s(F,D)+s(F,H)+s(G,D)+s(G,H))
- s(A,B)=0.8/6 (0.8+0+0+0+0+0)=0.8×0.8/6=0.64/6≈0.107

SimRank in Heterogeneous Bipartite Network

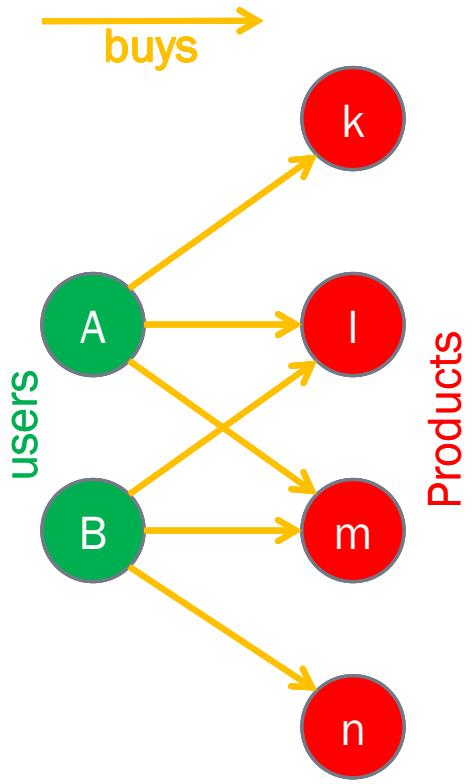
- In a heterogeneous network of users and products, the similarity of products and users are mutually-reinforced
  - two users can be considered similar if they buy similar products
  - two products can be considered similar if they are bought by similar users
- Similarity between two distinct users can be expressed as:

$$s(u_1, u_2) = \frac{C_1}{|O(u_1)|.|O(u_2)|} \sum_{i=1}^{|O(u_1)|} \sum_{j=1}^{|O(u_2)|} s(O_i(u_1), O_j(u_2))$$

- Similarity between two distinct products can be expressed as:

$$s(p_1, p_2) = \frac{C_2}{|I(p_1)|.|I(p_2)|} \sum_{i=1}^{|I(p_1)|} \sum_{j=1}^{|I(p_2)|} s(I_i(p_1), I_j(p_2))$$

Illustration: SimRank in Heterogeneous Bipartite Network



To calculate the similarity between users A and B

$O(A) = \{k, l, m\}$ and $O(B) = \{l, m, n\}$

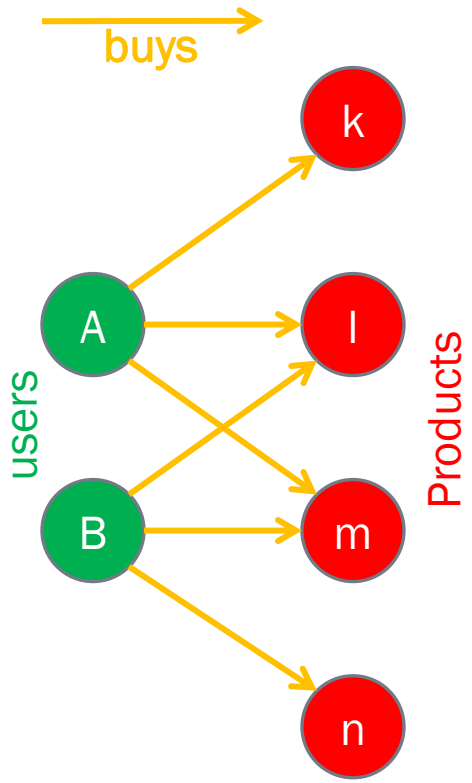
$I(k) = \{A\}$, $I(l) = \{A, B\}$, $I(m) = \{A, B\}$, and $I(n) = \{B\}$

$$s(A,B) = \frac{C_1}{3\times3}(s(k,l) + s(k,m) + s(k,n) + s(l,l) + s(l,m) + s(l,n) + s(m,l) + s(m,m) + s(m,n))$$

We have, $s(X,X) = 1$ and $s(X,Y) = s(Y,X)$

$$s(k,l) = \frac{C_2}{1\times2}[s(A,A) + s(A,B)] = \frac{C_2}{2} + \frac{C_2.s(A,B)}{2}$$

Illustration: SimRank in Heterogeneous Bipartite Network



Similarly, $s(k,m) = \frac{C_2}{2} + \frac{C_2.s(A,B)}{2}$, $s(k,n) = C_2.s(A,B)$

$s(l,l) = 1$, $s(l,m) = \frac{C_2}{2} + \frac{C_2.s(A,B)}{2}$, $s(l,n) = \frac{C_2}{2} + \frac{C_2.s(A,B)}{2}$

$s(m,l) = \frac{C_2}{2} + \frac{C_2.s(A,B)}{2}$, $s(m,m) = 1$, $s(m,n) = \frac{C_2}{2} + \frac{C_2.s(A,B)}{2}$

Solving, $s(A,B) = \frac{3C_1C_2+2C_1}{9-4C_1C_2}$

Further, setting $C_1 = C_2 = 0.8$,

$$\boldsymbol{s(A,B) = 0.547}$$

SimRank in Homogeneous Bipartite Network

Can you apply SimRank in following application ?

In a citation network, two scientific papers might be similar as survey papers if they cite similar result papers, while two papers might be similar as result papers if they are cited by similar survey papers.

References:

1) Social Network Analysis by Tanmoy Chakraborty

2) SimRank: a measure of structural-context similarity by Jeh, Glen and Widom, Jennifer, ACM, 2002. Link: https://dl.acm.org/doi/abs/10.1145/775047.775126